

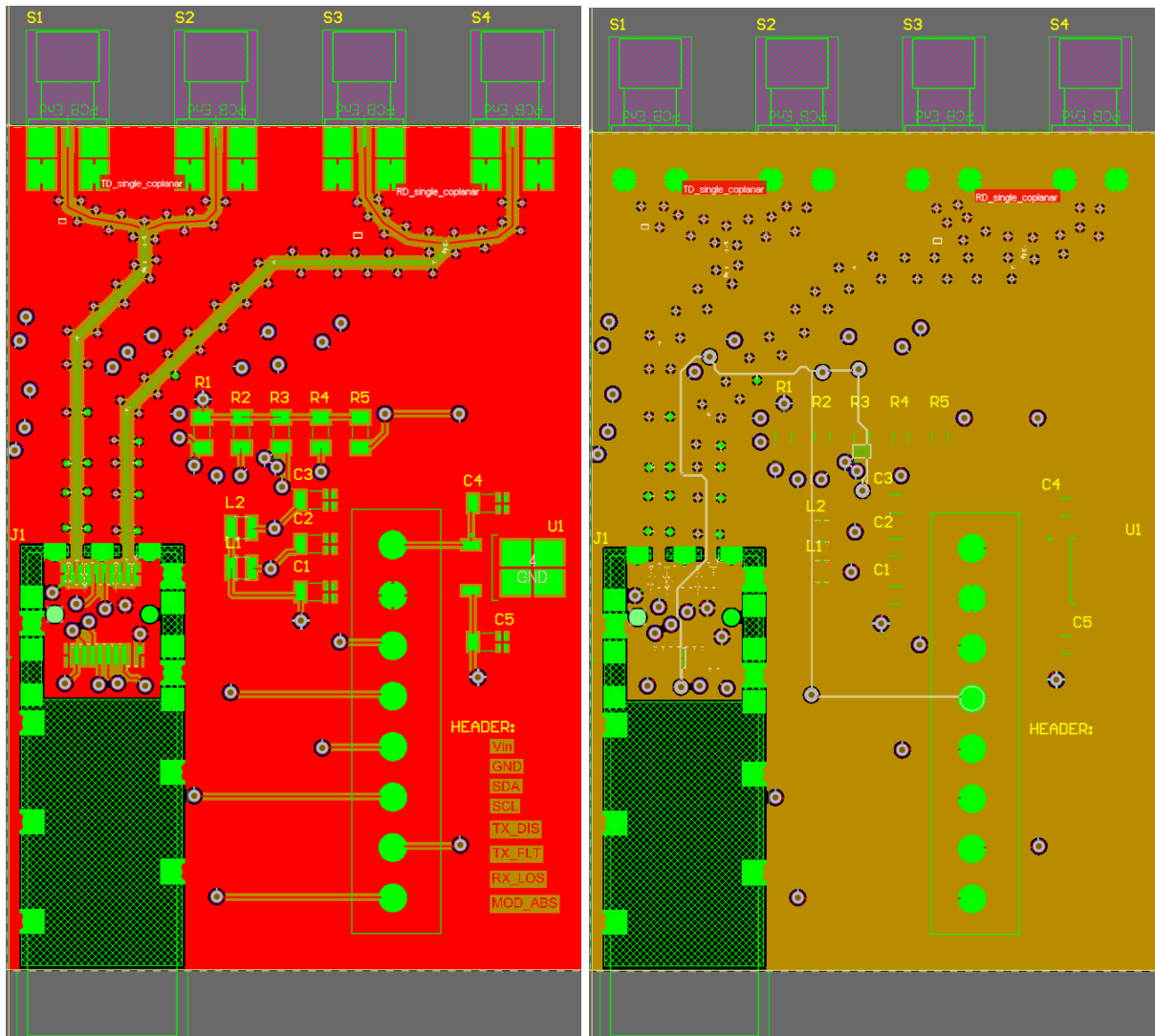


Figure 1: Top layer and Inner Layer 1

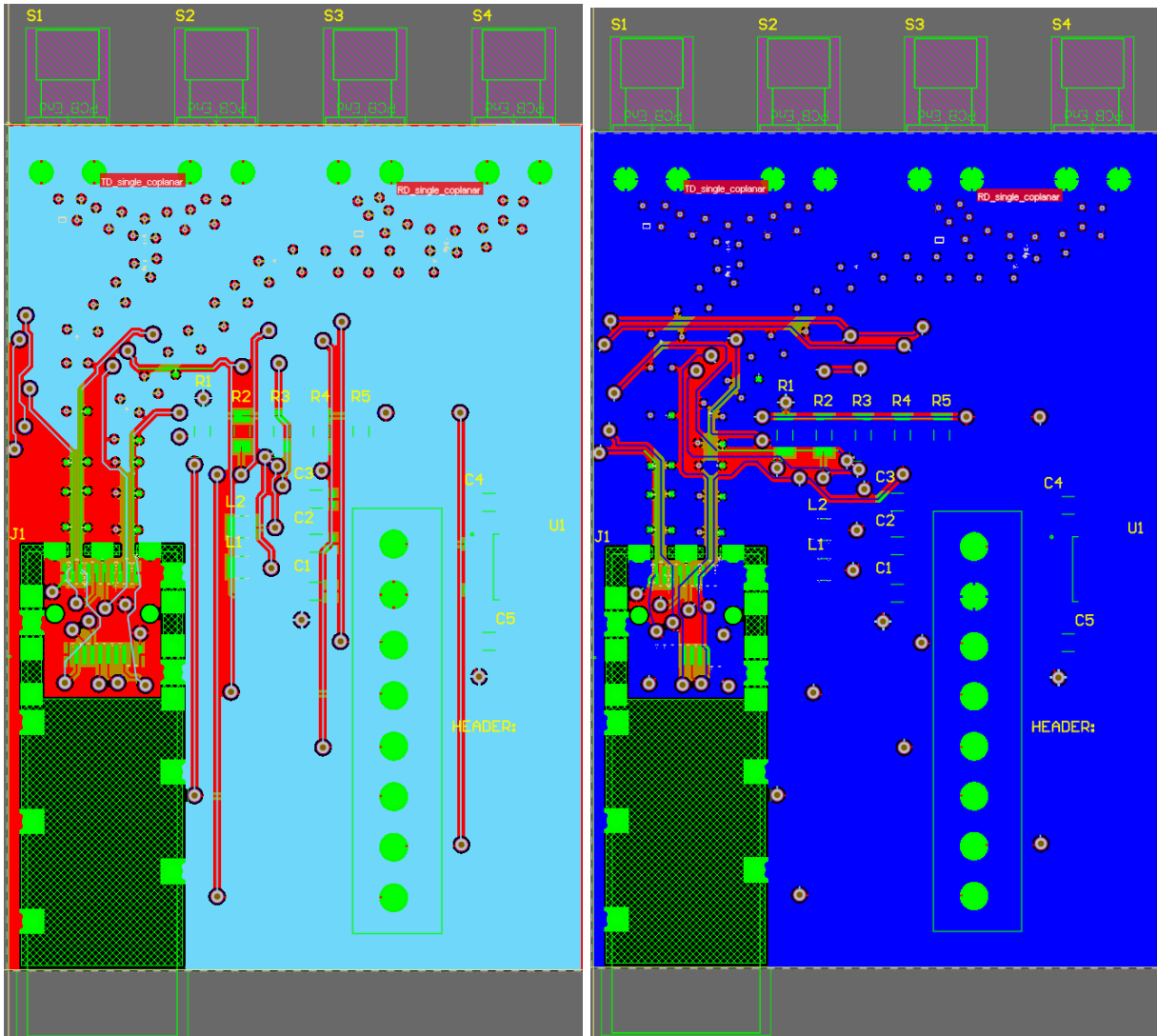


Figure 2: Inner layer 2 and Bottom layer

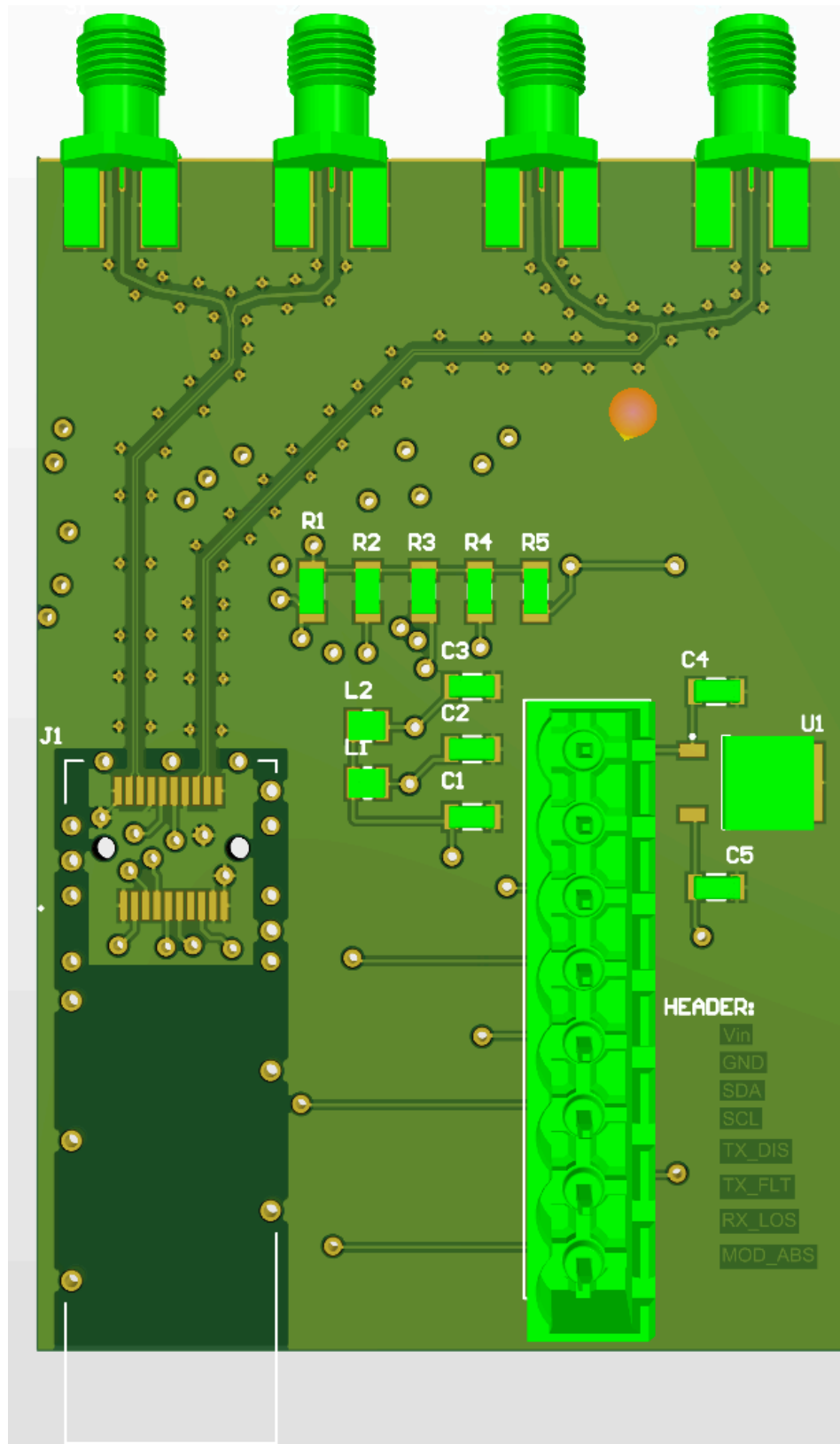


Figure 3: 3d representation

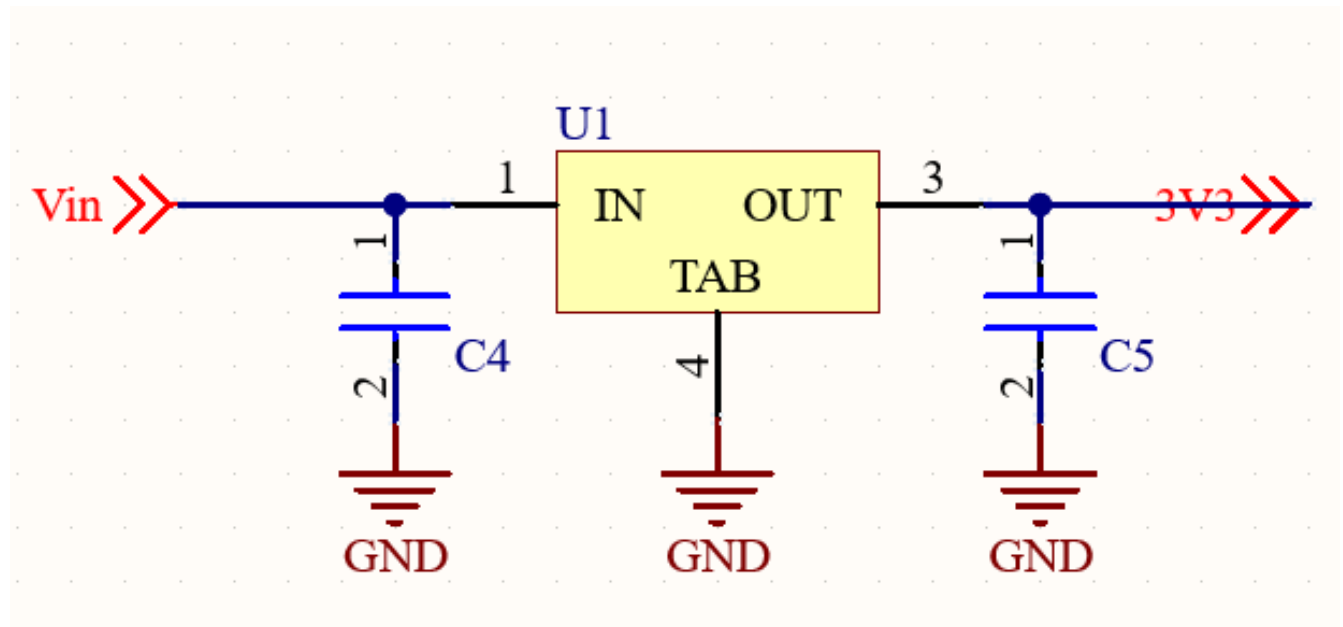


Figure 4: Voltage Regulator Schematic

#	Name	Material	Type	Weight	Thickness	Dk	Copper Orient...
	Top Overlay		Overlay				
	Top Solder	Solder Resist	Solder Mask		0.01016mm	3.5	
1	Top Layer		Signal	1oz	0.03556mm		Above
	Dielectric 2	PP-006	Prepreg		0.2104mm	4.4	
2	Inner layer 1	CF-004	Signal	1oz	0.0152mm		Above
	Dielectric 1	FR-4	Dielectric		1.065mm	4.6	
3	Inner Layer 2	CF-004	Signal	1oz	0.0152mm		Below
	Dielectric 3	PP-006	Prepreg		0.2104mm	4.4	
4	Bottom Layer		Signal	1oz	0.03556mm		Below
	Bottom Solder	Solder Resist	Solder Mask		0.01016mm	3.5	
	Bottom Overlay		Overlay				

Properties

Layer Stack Manager

Q

Search

Impedance Profile

Description

Type

Target Impedance

Target Tolerance

Single-Coplanar

50

10%

Board

Stack Symmetry

Library Compliance

Layers

Dielectrics

Conductive Thickness

Dielectric Thickness

Total Thickness

4

3

0.10152mm

1.4858mm

1.60764mm

Other

Roughness

Model Type

Surface Roughness (SR)

Roughness Factor (RF)

Copper Resistance

Flat Conductors

0um

1

17.24nΩ

Manufacturing Parameters

Via Plating Thickness

0.01524mm

Figure 5: Single-Coplanar impedance calculations

#	Name	Material	Type	Weight	Thickness	Dk	Copper Orient...	Top Ref	Bottom Ref	Width (W1)	Trace Gap...	Impe...	Deviation	Delay (Tp)	
	Top Overlay		Overlay												
	Top Solder	Solder Resist	Solder Mask		0.01016mm	3.5									
1	Top Layer		Signal	1oz	0.03556mm		Above	✓	2 - Inner lay...	0.14955mm	0.127mm	100.01	0.01%	5.49179ns/m	
	Dielectric 2	PP-006	Prepreg		0.2104mm	4.4									
2	Inner layer 1	CF-004	Signal	1oz	0.0152mm		Above	✓	1 - Top Layer	3 - Inner Lay...	0.09241mm	0.127mm	100.01	0.01%	7.1584ns/m
	Dielectric 1	FR-4	Dielectric		1.065mm	4.6									
3	Inner Layer 2	CF-004	Signal	1oz	0.0152mm		Below	✓	2 - Inner lay...	4 - Bottom L...	0.09241mm	0.127mm	100.01	0.01%	7.1584ns/m
	Dielectric 3	PP-006	Prepreg		0.2104mm	4.4									
4	Bottom Layer		Signal	1oz	0.03556mm		Below	✓	3 - Inner Lay...		0.14955mm	0.127mm	100.01	0.01%	5.49179ns/m
	Bottom Solder	Solder Resist	Solder Mask		0.01016mm	3.5									
	Bottom Overlay		Overlay												

Properties

Layer Stack Manager

Q Search

Impedance Profile

Description

Type

Target Impedance

Target Tolerance

Differential-Coplanar

100

10%

Board

Stack Symmetry

Library Compliance

Layers

Dielectrics

Conductive Thickness

Dielectric Thickness

Total Thickness

4

3

0.10152mm

1.4858mm

1.60764mm

Other

Roughness

Model Type

Surface Roughness (SR)

Roughness Factor (RF)

Copper Resistance

Flat Conductors

0um

1

17.24nΩ

Manufacturing Parameters

Via Plating Thickness

0.01524mm

Figure 6: Differential-Coplanar impedance calculations